Multi-Objective Online Learning

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References

Chapter 3 of Aaron's lecture notes:
https://www.cis.upenn.edu/~aaroth/GamesInLearning.pdf

Online Learning with Multiple Subsequences

For $t = 1, 2, \dots, T$:

Learner observes a context $C_t \in \{0,1\}^K$

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss vector $\ell_t = \left(\ell_t(1), \dots, \ell_t(N)\right) \in \mathbb{R}^N$

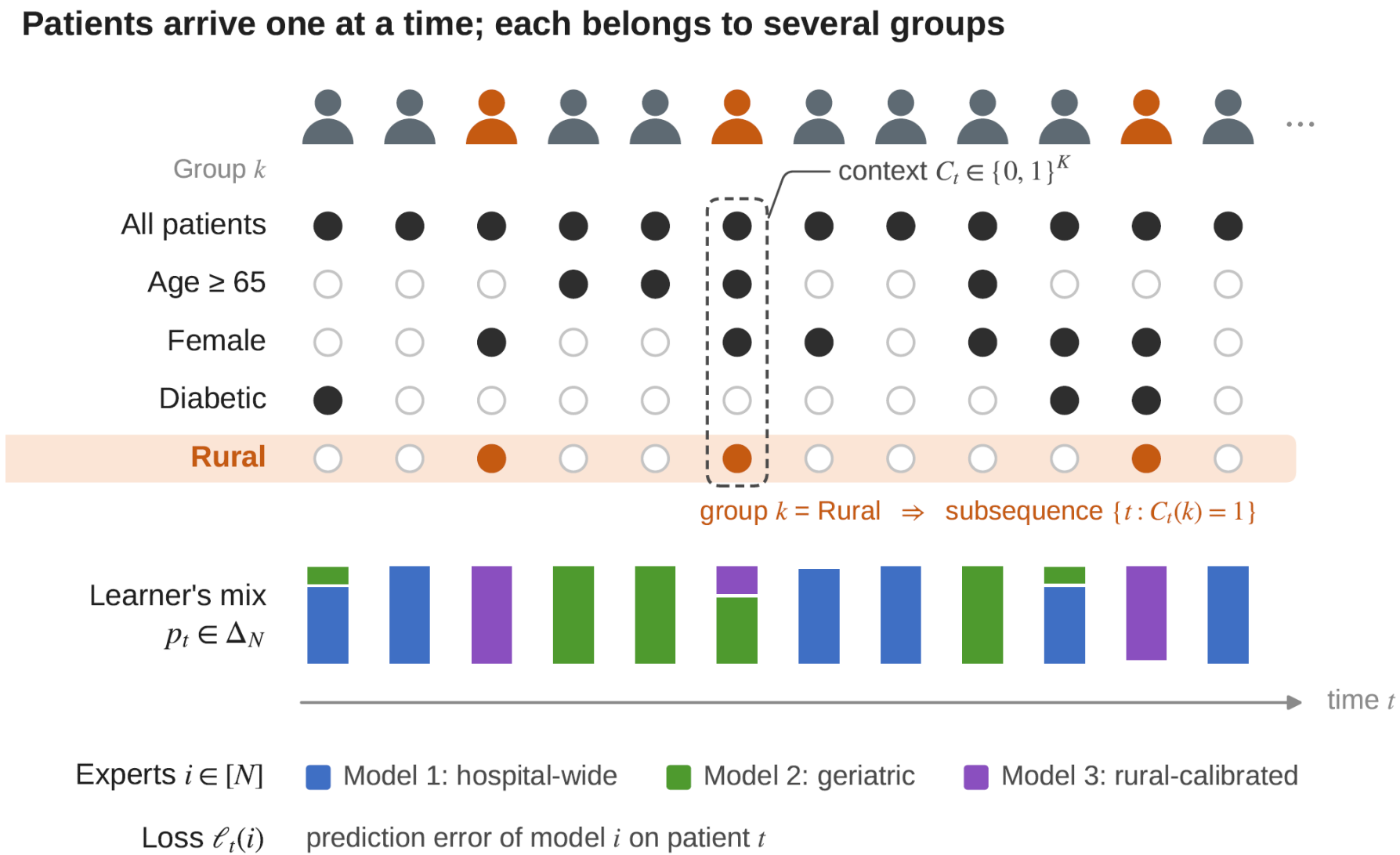
Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

$$\text{Group-wise Regret} = \max_{k \in [K]} \left(\sum_{t:\, C_t(k)=1} \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t:\, C_t(k)=1} \ell_t(i) \right)$$

$$= \max_{k \in [K]} \max_{i \in [N]} \sum_{t=1}^{T} C_t(k) \left(\langle p_t, \ell_t \rangle - \ell_t(i) \right)$$

Online Learning with Multiple Subsequences

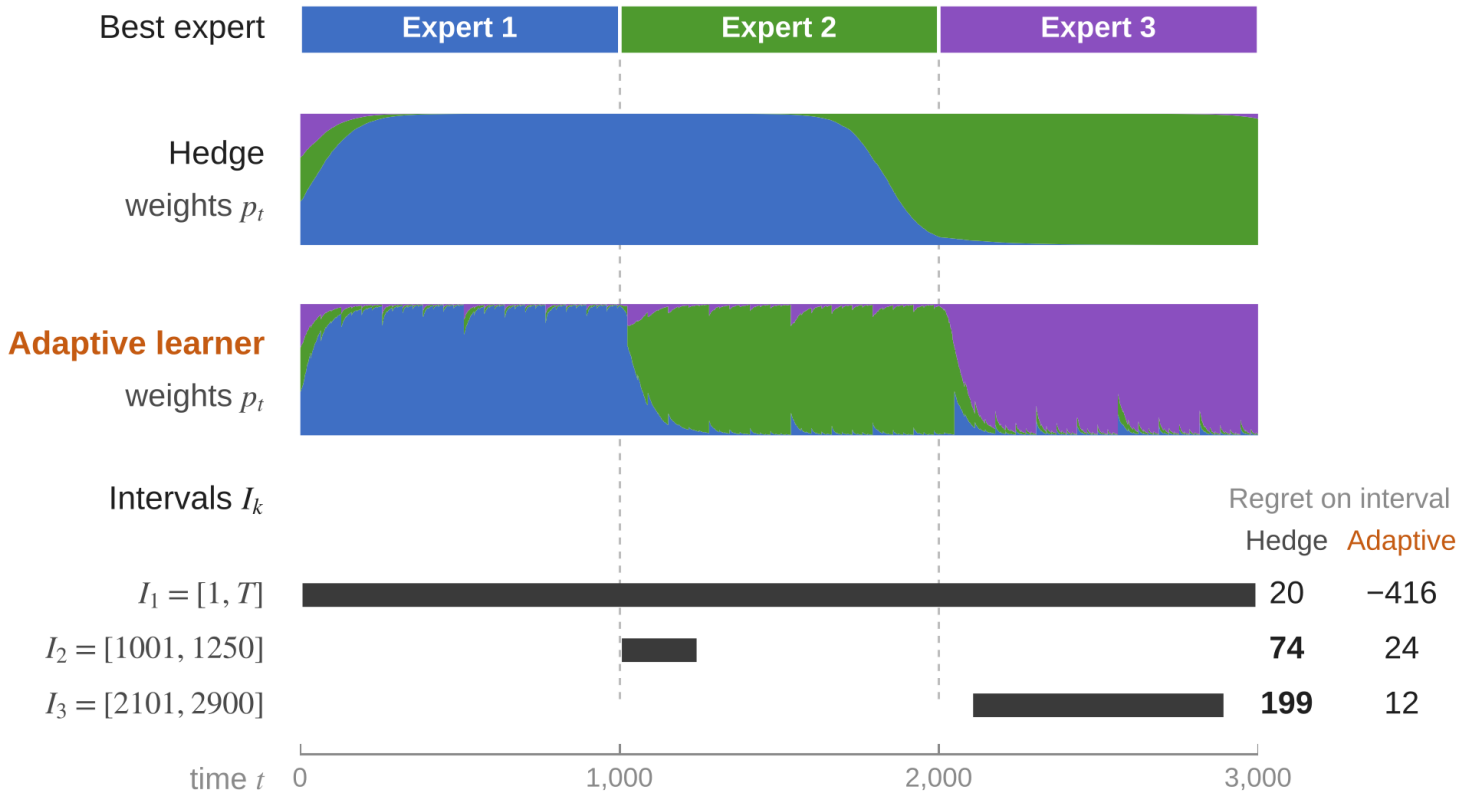


Online Learning with Multiple Subsequences

(a) The best expert changes over time; every interval is a group

Adaptive regret $= \max_{I=[s,e]\subseteq[T]} \left(\sum_{t\in I} \langle p_t, \ell_t \rangle - \min_{i\in[N]} \sum_{t\in I} \ell_t(i) \right)$

= group-wise regret with one group per interval: $C_t(k) = \mathbf{1}\{t \in I_k\}$



Is the standard no-regret guarantee sufficient?

Our goal

$$\max_{k\in[K]} \max_{i\in[N]} \sum_{t=1}^{T} C_t(k)(\langle p_t, \ell_t \rangle - \ell_t(i)) \leq O\left(\sqrt{T \log(NK)}\right)$$

In fact, the following stronger guarantee is possible:

$$\forall k \in [K] \qquad \max_{i\in[N]} \sum_{t=1}^{T} C_t(k)(\langle p_t, \ell_t \rangle - \ell_t(i)) \leq O\left(\sqrt{\left(\sum_{t=1}^{T} C_t(k)\right) \log(NK)}\right)$$

Recall: No-Regret Best-Response Dynamics

With a known $G \in \mathbb{R}^{M \times N}$

p-player (minimizer)

For $t = 1,2, \ldots$

Choose $p_t = \underset{p \in \Delta_M}{\operatorname{argmin}}\, p^\top G q_t$

q-player (maximizer)

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $G^\top p_t \in \mathbb{R}^N$

Boosting:	Model player	Sample player
LP:	Variable player	Constraint player

Recall: No-Regret Best-Response Dynamics

Guarantee:

$$\max_{q} \sum_{t=1}^{T} p_t^{\top} G q \leq T \cdot \text{GameValue}(G) + O\left(\sqrt{T \log N}\right)$$

Modified Scenario

With a time-varying and unknown $G_t \in \mathbb{R}^{M \times N}$

Additional assumption:

$$\forall q \ \exists p \ \forall G_t \quad p^\top G_t q \leq V^\star$$

p-player (minimizer)

For $t = 1,2, \ldots$

~~Choose $p_t = \underset{p \in \Delta_M}{\operatorname{argmin}} \, p^\top G q_t$~~

Choose p_t such that $p_t^\top G_t q_t \leq V^\star$

q-player (maximizer)

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $G_t^\top p_t \in \mathbb{R}^N$

Modified Scenario

Guarantee:

$$\max_{q} \sum_{t=1}^{T} p_t^{\top} G_t q \leq TV^{\star} + O\left(\sqrt{T \log N}\right)$$

A “trivial” instantiation of the dynamics

$G_t \in \mathbb{R}^{N \times N}$

$G_t(i,j) = \ell_t(i) - \ell_t(j)$

Verify: $\forall q \ \exists p \ \forall G_t \quad p^\top G_t q \leq 0$

Guarantee:

A “trivial” instantiation of the dynamics

$G_t \in \mathbb{R}^{N \times N}$

$G_t(i,j) = \ell_t(i) - \ell_t(j)$

The interaction just looks like below (somewhat trivial):

p-player (minimizer)

For $t = 1,2, \ldots$

Choose $p_t = q_t$

q-player (maximizer)

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $r_t(j) = \langle p_t, \ell_t \rangle - \ell_t(j)$

Back to "Online Learning with Multiple Subsequences"

For $t = 1, 2, \ldots, T$:

Learner observes a context $C_t \in \{0,1\}^K$

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss vector $\ell_t = \left(\ell_t(1), \ldots, \ell_t(N)\right) \in \mathbb{R}^N$

Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

$$\text{Group-wise Regret} = \max_{k \in [K]} \left(\sum_{t:\, C_t(k)=1} \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t:\, C_t(k)=1} \ell_t(i) \right)$$

$$= \max_{k \in [K]} \max_{i \in [N]} \sum_{t=1}^{T} C_t(k) \left(\langle p_t, \ell_t \rangle - \ell_t(i) \right)$$

Back to “Online Learning with Multiple Subsequences”

$G_t \in \mathbb{R}^{N \times NK}$

$G_t(i,(j,k)) = C_t(k)(\ell_t(i) - \ell_t(j))$

Verify: $\forall q, C_t \ \exists p \ \forall \ell_t \qquad p^\top G_t q \leq 0$

Back to “Online Learning with Multiple Subsequences”

$G_t \in \mathbb{R}^{N \times NK}$

$G_t(i,(j,k)) = C_t(k)(\ell_t(i) - \ell_t(j))$

Guarantee:

Resulting Algorithm

For $t = 1,2, \ldots$

Choose $p_t \in \Delta_N$ as

$$p_t(i) = \frac{\sum_k q_t(i,k) C_t(k)}{\sum_{j,k} q_t(j,k) C_t(k)}$$

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_{NK}$ from no-regret alg

Feed no-regret alg with reward

$r_t(j,k) = C_t(k)(\langle p_t, \ell_t \rangle - \ell_t(j))$